SPECIFICATION

Cross-Reference to Related Applications

This application is related to the inventions described in the U.S. Patent Applications for "A Method and System for High-Radix Physical Layer Data Encoding and Representation" and "A Protocol Stack and System for Parametric Data Networking." The present invention applies these core principles to the uniquely challenging and dynamic environment of troposcatter communication, the contents of which are hereby incorporated by reference in their entirety.

Background of the Invention

Tropospheric scatter (troposcatter) communication provides a valuable capability for establishing beyond-line-of-sight (BLOS) links, particularly in military and remote operational contexts. However, the tropospheric channel is characterized by severe and rapid fading, multipath distortion, and high path loss, making it notoriously difficult. Conventional troposcatter systems focus on achieving a robust *binary* data link through powerful error correction, diversity schemes, and complex modulation techniques. These systems are fundamentally limited to transporting bit-streams, meaning any high-precision or symbolic data must first be converted to a lossy, approximated binary format before transmission.

Summary of the Invention

The present invention defines an adaptive troposcatter communication system that natively transmits high-radix symbolic data, overcoming the limitations of binary systems and preserving perfect data precision across the link. The system's core innovation is its ability to adapt its physical layer encoding to the volatile nature of the troposcatter channel.

The system comprises a pair of adaptive transceivers. Each transceiver includes a channel condition monitoring unit that continuously analyzes key channel metrics (e.g., signal-to-noise ratio, fading characteristics). A dynamic symbol-to-state mapper uses this real-time data to define and continuously update an optimal set of N distinguishable RF states for transmission. The number of states, N (the radix), is not fixed; the mapper can increase N when the channel is clear to boost data rate, and decrease N to a smaller, more robust set of states when the channel is degraded, thereby ensuring link reliability.

The system's modulator directly converts abstract symbols (representing numbers, operators, or other data primitives) into this dynamically defined set of RF states. Data is encapsulated in parametric packets, allowing unevaluated mathematical "recipes" to be transmitted as the payload. This ensures that even when the radix N is dynamically changed, the symbolic integrity of the underlying data is preserved from end to end.

Brief Description of the Drawings

FIG. 1 is a conceptual diagram illustrating the adaptive high-radix parametric data flow in a troposcatter communication system.

FIG. 2 is a block diagram of an adaptive troposcatter transceiver highlighting the dynamic symbol-to-state mapping based on channel monitoring.

Detailed Description of the Invention

Referring to **FIG. 1**, the adaptive troposcatter system (100) is shown. A first adaptive transceiver (130) receives a parametric packet (120) containing a symbolic data structure (110). A channel monitoring unit within the transceiver assesses the volatile troposcatter path (140). Based on this assessment, an adaptive mapper (135) selects an optimal set of N RF states for encoding. The transceiver then transmits a high-radix RF signal (145) which is scattered by the troposphere. A second adaptive transceiver (150) receives the scattered signal. Its own adaptive mapper (155), using its own channel assessment, decodes the signal by selecting the correct inverse mapping. This allows the reconstruction of the original parametric packet (120), delivering the perfect-precision symbolic data (110) to the destination.

FIG. 2 provides a block diagram of the adaptive transceiver (200). A Channel Condition Monitor (210) continuously analyzes the received signal from the antenna (250). It sends metrics to a Dynamic Symbol-to-State Mapper (220). This mapper defines the optimal radix N and the specific mapping of symbols to RF states. It provides this map to both the High-Radix RF Modulator (230), which encodes outgoing abstract symbols, and the High-Radix RF Demodulator (240), which decodes incoming RF signals. This closed-loop adaptation ensures the most efficient and reliable radix is used at all times.